

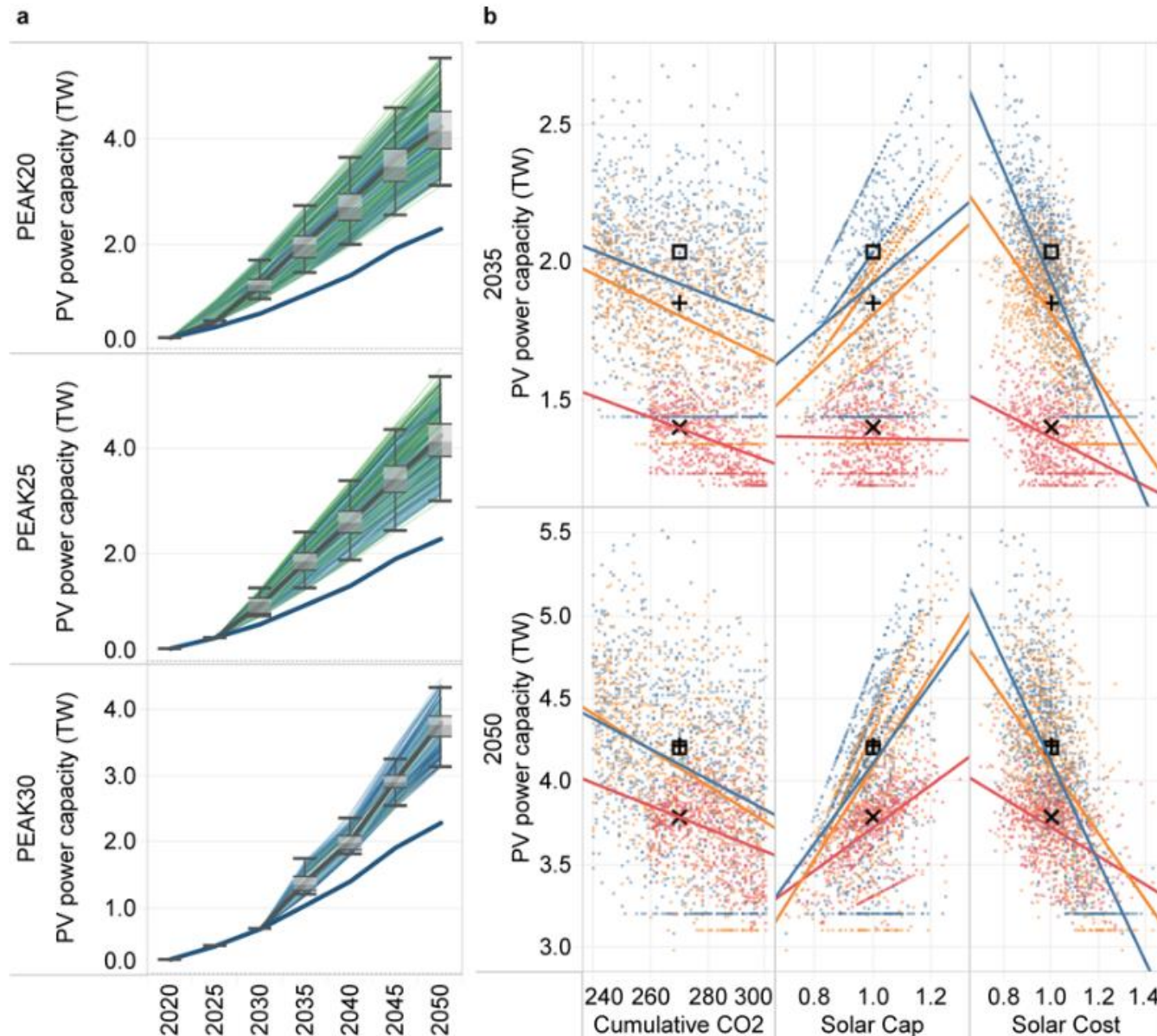
GCAM-HPC-PKU

基于计算集群的GCAM工具

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Modelling scenario ensemble



A large scenario ensemble is needed to capture multidimensional **uncertainties**, quantify the range and likelihood of possible transition pathways, and identify **robust strategies** across diverse futures.

Socio-economy (SSPs)

middle-of-the-road; taking-the-green-road
a-rocky-road; a-road-divided; taking-the-highway



Radiative forcing (PCRs)

RCP2.6; RCP4.5; RCP6.0; RCP8.5



Policy innervation

Carbon tax; Carbon Constraint;
Nation Determined Commitments (NDC)



Technological potential

High CCS; High Nuclear; High Renewables;

Thousands of scenarios



GCAM辅助工具 – 高性能计算

gcam-hpc-PKU Public

main

Go to file

Code

jerrysong0128 and codex Update tutorial deck f... 30b99aa · yesterday 44 Commits

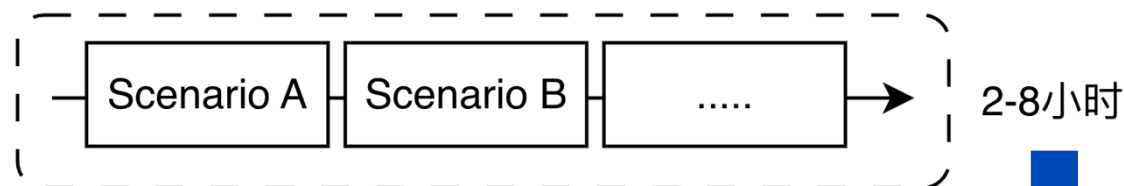
asset/tutorial	Update tutorial deck for the canonic...	yesterday
gcam-hpc-tools	Document manual GCAM upload and...	yesterday
gcam-runjob-log	Keep runtime directory structure wit...	yesterday
gcam-scratch	Keep runtime director	
.gitattributes	Track gcam-hpc-pku-	
.gitignore	Document manual GC	
LICENSE.md	add ECL-2.0 LICENSE	
README.md	Add the parallel RUN	

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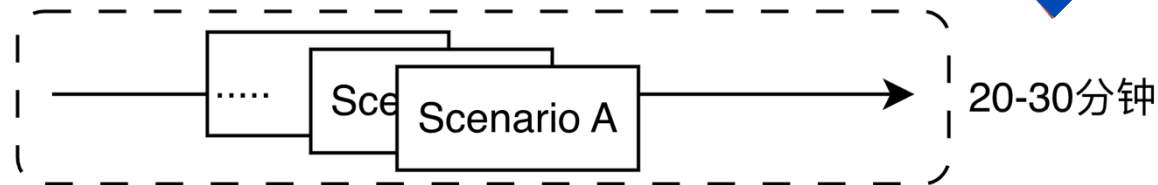
<https://github.com/jerrysong0128/gcam-hpc-PKU>

多情景并行，实现规模化测试及情景阵列构建
(适配任意版本GCAM-core/GCAM-China)

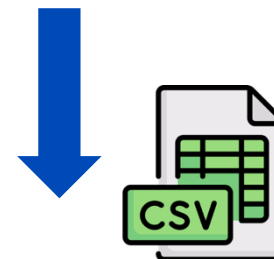
顺序计算求解：



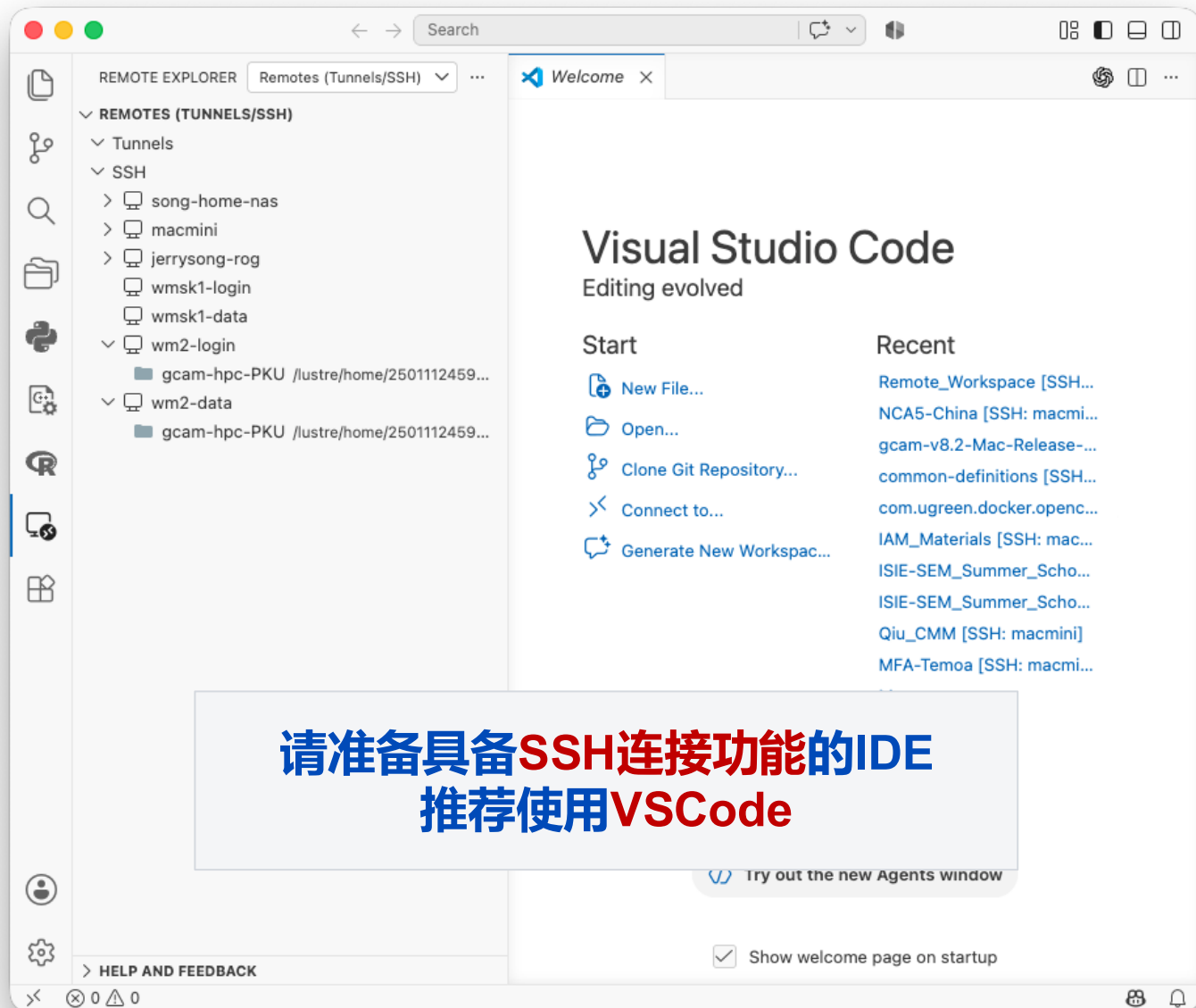
高性能并行计算求解：



服务器端**定制处理**模型结果
合并为**轻量级csv格式**



0--IDE准备：VSCode与超算链接

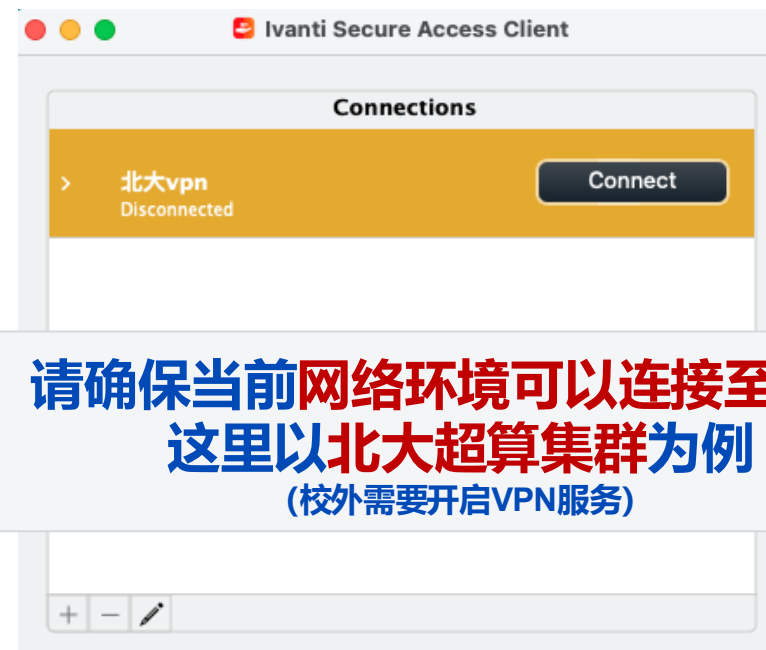


https://its.pku.edu.cn/service_1_vpn.jsp



VPN 服务

为方便北京大学校园网用户在校外（家中、出差或国外）访问校园网资源，计算中心提供了VPN服务，可安全地接入校园网，如同在校内一样方便地访问学校全部的内网资源与服务（如校内门户、电子期刊数据资源等）。



0--gcam-hpc-PKU仓库目录结构

所有脚本围绕以下目录组织，理解目录职责是理解操作流程的前提。

```
gcam-hpc-PKU/
├─ gcam-core/ 或 gcam-china-*/      GCAM 源码与编译产物
├─ gcam-hpc-tools/
│   ├── build-tools/                environment.sh · build 依赖包 · profiles
│   ├── run-tools/                  RUN、QUERY 执行与结果合并脚本
│   ├── query-tools/                QUERY 生成器与本地用户 QUERY XML
│   ├── local-tools/                本地后处理Query CSV工具
│   ├── configuration-sets/         配置模板与 batch XML
│   └─ run-pipeline.sh              同步 → 情景展开 → Slurm 提交 → 合并
├─ gcam-runjob-log/                 Slurm 日志
└─ gcam-scratch/                    运行输入与输出
```

1-BUILD 编译

利用GCAM源代码，编译linux版本的gcam.exe; 解压 HPC 依赖，统一加载 environment.sh 后编译。

2-RUN 运行

run-pipeline.sh 同步输入与预配置 QUERY，生成情景和 Slurm 作业；每个情景运行后执行同一组查询。

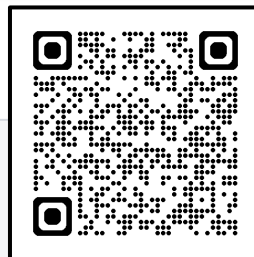
3-QUERY 查询逻辑

查询 XML 须提前写好；所有情景查询 CSV 最终自动合并为一个文件。

1--BUILD: 克隆仓库与准备编译工具

1 · 克隆工具仓库

```
git clone https://github.com/jerrysong0128/gcam-hpc-PKU.git
cd gcam-hpc-PKU
```



2 · 本地下载 GCAM 源代码并上传超算

```
gcam-core: github.com/JGCRI/gcam-core/releases
gcam-china: github.com/umd-cgs/gcam-china/releases
# 上传并解压到仓库根目录；源码须包含 cvs/objects/build/linux
```

3 · 解压依赖并加载统一环境

```
cd gcam-hpc-tools/build-tools
unzip gcam-hpc-pku-libraries.zip
unzip archive/libraries.zip          # 生成 libs/

export GCAM_HPC_WORKSPACE=/absolute/path/to/gcam-hpc-PKU
export GCAM_HPC_CLUSTER=wm2
source environment.sh
```

libs

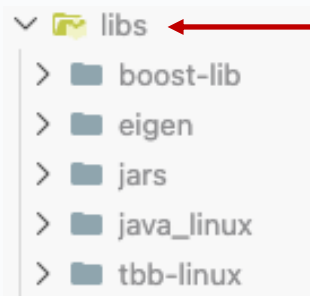
- > boost-lib
- > eigen
- > jars
- > java_linux
- > tbb-linux

environment.sh 自动识别 gcam-core / gcam-china 源码目录，并加载 profiles/<cluster>.profile。

1--BUILD: 编译生成 gcam.exe

依赖目录

```
cd gcam-hpc-tools/build-tools
unzip gcam-hpc-pku-libraries.zip
unzip archive/libs.zip
```



需要参考wm2.profile, 修改对应HPC的路径, 指向准备好的libs

加载环境并编译

```
export GCAM_HPC_WORKSPACE=/path/to/gcam-hpc-PKU
export GCAM_HPC_CLUSTER=wm2
source gcam-hpc-tools/build-tools/environment.sh
cd "$GCAM_LIB"
make clean
make gcam -j 16
```

编译成功标志

编译退出码为 0 · exe/gcam.exe 已生成

集群 profile build-tools/profiles/wm2.profile

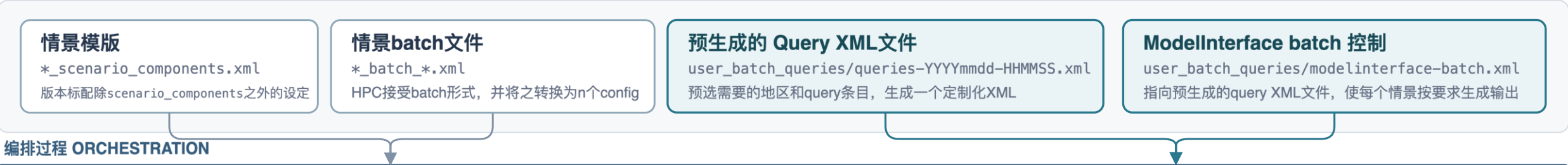
```
1 # Purpose: Define compiler, local libraries, Java, and Slurm for WEIMING-2.
2 # Author: Jingyang Song, Peking University; Jul 2026;
3 #
4 # wm2.profile -- WEIMING-2 cluster
5 # Sourced by environment.sh. Do not source directly.
6 # Provides build env (compiler, BOOST/JAVA/TBB/EIGEN, JARS) and SLURM_TEMPLATE.
7
8 module load gcc/12.2.0
9 export CXX=g++
10
11 _GCAM_LIBS="${TOOLDIR}/build-tools/libs"
12
13 export BOOST_INCLUDE="${_GCAM_LIBS}/boost-lib"
14 export BOOST_LIB="${_GCAM_LIBS}/boost-lib/lib"
15
16 export JAVA_HOME="${_GCAM_LIBS}/java_linux"
17 export JAVA_INCLUDE="${JAVA_HOME}/include"
18 export JAVA_LIB="${JAVA_HOME}/lib/server"
19 export JARS_LIB="${_GCAM_LIBS}/jars/*"
20
21 export EIGEN_INCLUDE="${_GCAM_LIBS}/eigen"
22
23 export TBB_INCLUDE="${_GCAM_LIBS}/tbb-linux/include"
24 export TBB_LIB="${_GCAM_LIBS}/tbb-linux/lib/intel64/gcc4.8"
25
26 export SLURM_TEMPLATE="${TOOLDIR}/run-tools/slurm-templates/wm2-run-template.slurm"
27
28 unset _GCAM_LIBS
29
```

以北京大学“未明二号”(HPC)为例

2--RUN: 工作逻辑 (optional)

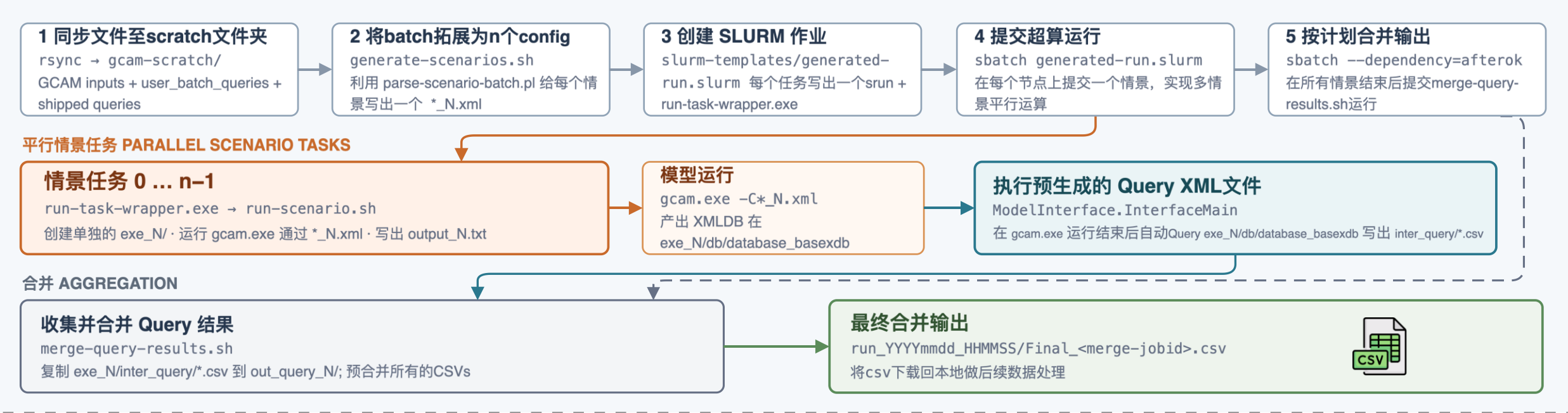
GCAM-HPC-PKU: RUN + QUERY

输入设定 PRE-RUN CONFIGURATION



编排过程 ORCHESTRATION

run-pipeline.sh 创造运行环境 · 同步inputs和query XML · 拓展情景 · 创建 SLURM 作业 · 提交超算运行 · 合并输出



2--RUN: 用 run-pipeline.sh 提交作业

1 · 运行前检查: QUERY 已配置

```
test -f gcam-hpc-tools/query-tools/user_batch_queries/modelinterface-batch.xml  
# 确认 <queryFile> 指向已生成的 queries-*.xml
```

2 · source 环境并提交运行

```
"PATH-TO-GCAM-HPC-PKU/gcam-hpc-tools/run-pipeline.sh" \           版本标配除scenario_components之外的设定  
"PATH-TO-GCAM-HPC-PKU/gcam-hpc-tools/configuration-sets/config/v82_default/v82_default_scenario_components.xml" \  
"PATH-TO-GCAM-HPC-PKU/gcam-hpc-tools/configuration-sets/config/v82_default/v82_default_batch_2.xml" 情景组合batch文件
```

3 · run-pipeline.sh 交互式确认

- ① 是否生成情景排列 (permutations)
- ② 同时运行多少个任务 (并行 task 数)
- ③ 是否立即用 sbatch 提交

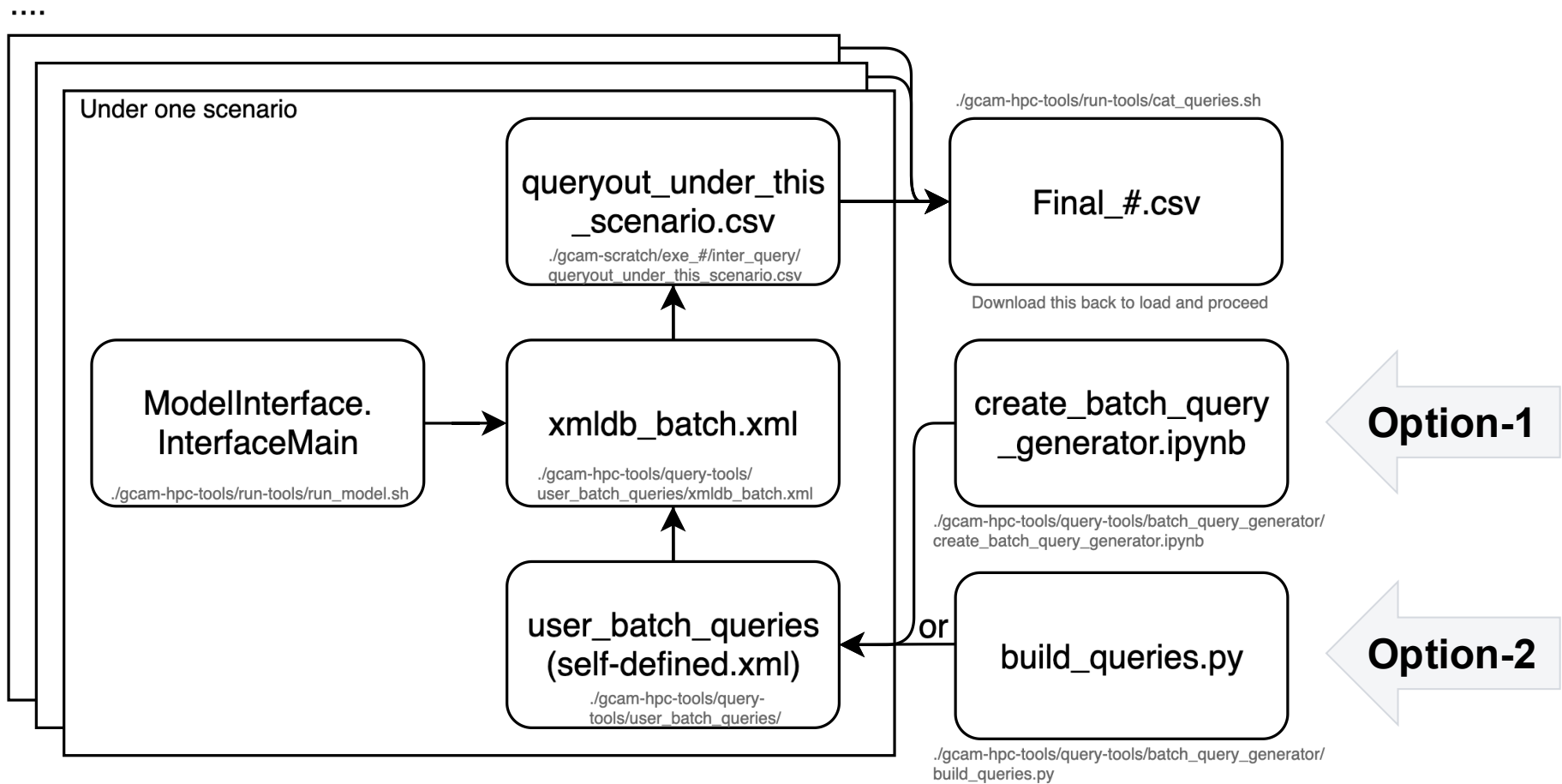
4 · 查看运行状态

```
squeue -u $(whoami)  
bash gcam-hpc-tools/run-tools/monitor-jobs.sh
```

3--QUERY: 预生成 Query XML文件

1 · 生成用户批量查询

提交 RUN 前先生成查询 XML，并让 xmldb_batch.xml 指向它。运行时，每个 run_model.sh 任务在 gcam.exe 结束后调用 ModelInterface，执行这组预配置查询并导出 CSV。



3--QUERY: 预生成 Query XML 并接入运行流程

1 · 生成用户批量查询 (Option-1)

query-tools/batch_query_generator/interactive-query-builder.ipynb # Notebook

The screenshot shows a Jupyter Notebook interface. At the top, a code cell contains the following Python code:

```
regions = create_batch_query_tools.read_regions_from_csv(gcam_core_path+"/input/gcamdata/inst/extdata/common/GCAM_region_names.csv")
queries = gcamreader.parse_batch_query(gcam_core_path+"/output/queries/Main_queries.xml")
result = create_batch_query_tools.select_queries_and_regions(queries, regions)# start selection panel
```

Below the code cell, the output is displayed. It consists of two side-by-side selection panels. The left panel is titled "Regions:" and lists the following options: All, USA, Africa_Eastern, Africa_Northern, Africa_Southern, Africa_Western, Australia_NZ, Brazil, Canada, Central America and Caribbean, Central Asia, China, EU-12, EU-15, and Ukraine. The right panel is titled "Queries:" and lists the following options: All, primary energy consumption by region (avg fossil efficiency), primary energy consumption by region (direct equivalent), primary energy consumption with CCS by region (direct equivalent), resource production, resource production by tech and vintage, resource supply curves, regional primary energy prices, elec gen by region (incl CHP), elec gen by subsector, elec gen by gen tech, elec gen by gen tech and cooling tech, elec gen by gen tech and cooling tech and vintage, elec gen by gen tech and cooling tech (new), and elec energy input by subsector. Below the "Regions:" panel, there is a green button labeled "Confirm Selection".

交互式 QUERY 生成器

2 · 配置 modelinterface-batch.xml

run-pipeline.sh 将 user_batch_queries/ 同步到 scratch/output/xmlldb_queries/; <queryFile> 必须指向同步后的文件。

```
<queryFile>../output/xmlldb_queries/queries-20260726-110822.xml</queryFile>
```

3--QUERY: 预生成 Query XML 并接入运行流程

1 · 生成用户批量查询 (Option-2)

```
query-tools/batch_query_generator/query-config.yaml #Query XML settings
```

```
# GCAM batch query builder config
#
# How to use:
# 1. Copy region names from available_regions into selected_regions.
# 2. Copy query titles from available_queries into selected_queries.
# 3. Run: python build_queries.py query-config.yaml
#
# Use All to select every available region or query.
```

```
selected_regions:
```

```
- All
```

```
selected_queries:
```

```
- iron and steel production by region
- iron and steel production by tech
```

```
query-tools/batch_query_generator/build_queries.py #Create Query XML with script
```

```
• (base) jerrysong@bogon Remote_Workspace % cd /Users/jerrysong/Study_PKU/Projects/Remote_Workspace/gcam-hpc-PKU/gcam-hpc-tools/query-tools/batch_query_generator
• (base) jerrysong@bogon batch_query_generator % python build_queries.py query-config.yaml
Generated: /Users/jerrysong/Study_PKU/Projects/Remote_Workspace/gcam-hpc-PKU/gcam-hpc-tools/query-tools/user_batch_queries/queries-20260727-220824.xml
Queries: 2
Regions: 32
```

2 · 配置 modelinterface-batch.xml

run-pipeline.sh 将 user_batch_queries/ 同步到 scratch/output/xmlldb_queries/; <queryFile> 必须指向同步后的文件。

```
<queryFile>../output/xmlldb_queries/queries-20260726-110822.xml</queryFile>
```

4--OUTPUT: 输出位置

输出目录

```
gcam-runjob-log/  
├─ gcamCatCSV.<jobid>.out  
└─ gcamRunjob.<jobid>.out
```

各作业输出 Pipeline合并过程
各作业输出 Pipeline运行过程

run-pipeline.sh 相关的error

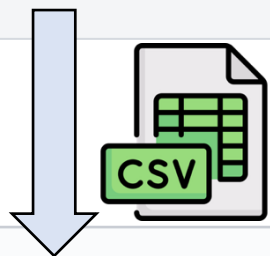
单情景gcam运行相关的error

```
gcam-scratch/output/run_YYYYmmdd_HHMMSS/  
├─ output_<task>.txt  
├─ out_query_<task>/  
└─ Final_<merge-jobid>.csv
```

各任务标准输出
各任务 QUERY CSV
merge-query-results.sh 合并结果

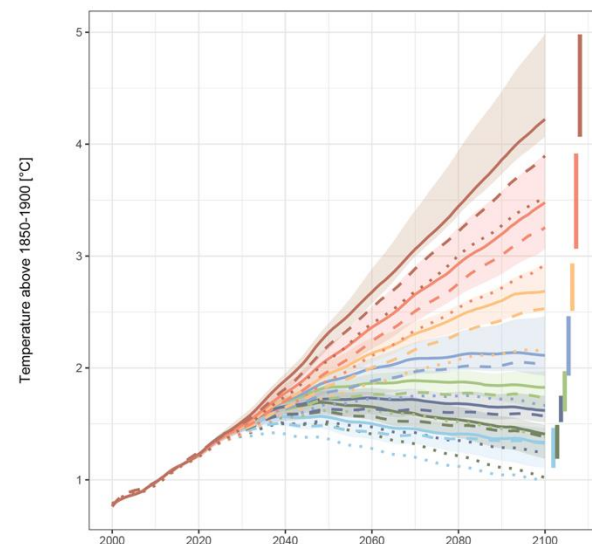
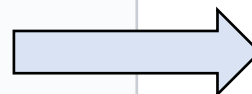
单情景gcam运行结果

多情景gcam结果合并

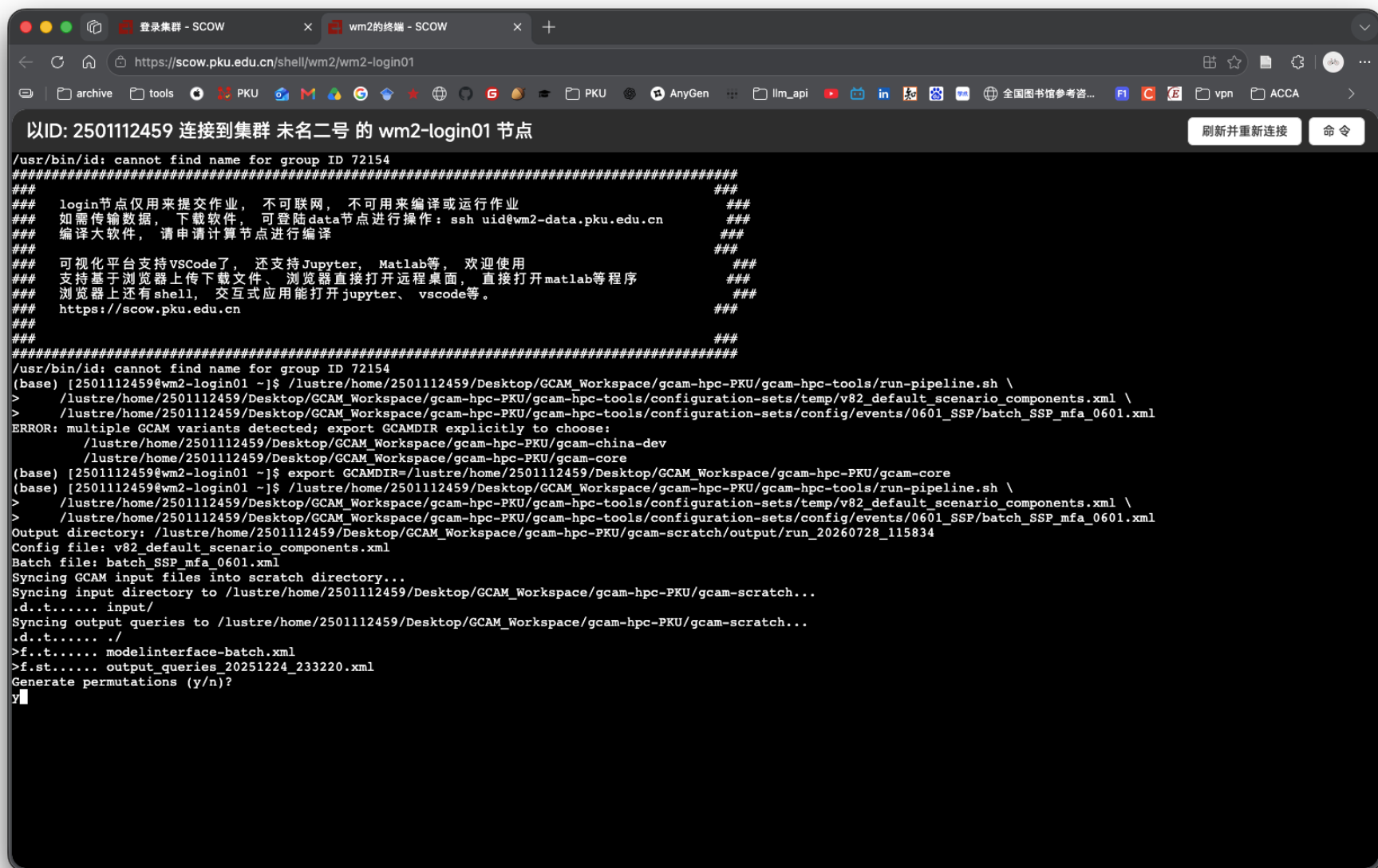


本地做后处理

```
gcam-hpc-PKU/gcam-hpc-tools/local-tools/  
├─ data_prepare_main_HPC.R  
├─ diag_header_HPC.R  
└─ diag_parser_HPC.R
```



2--RUN: Demonstration

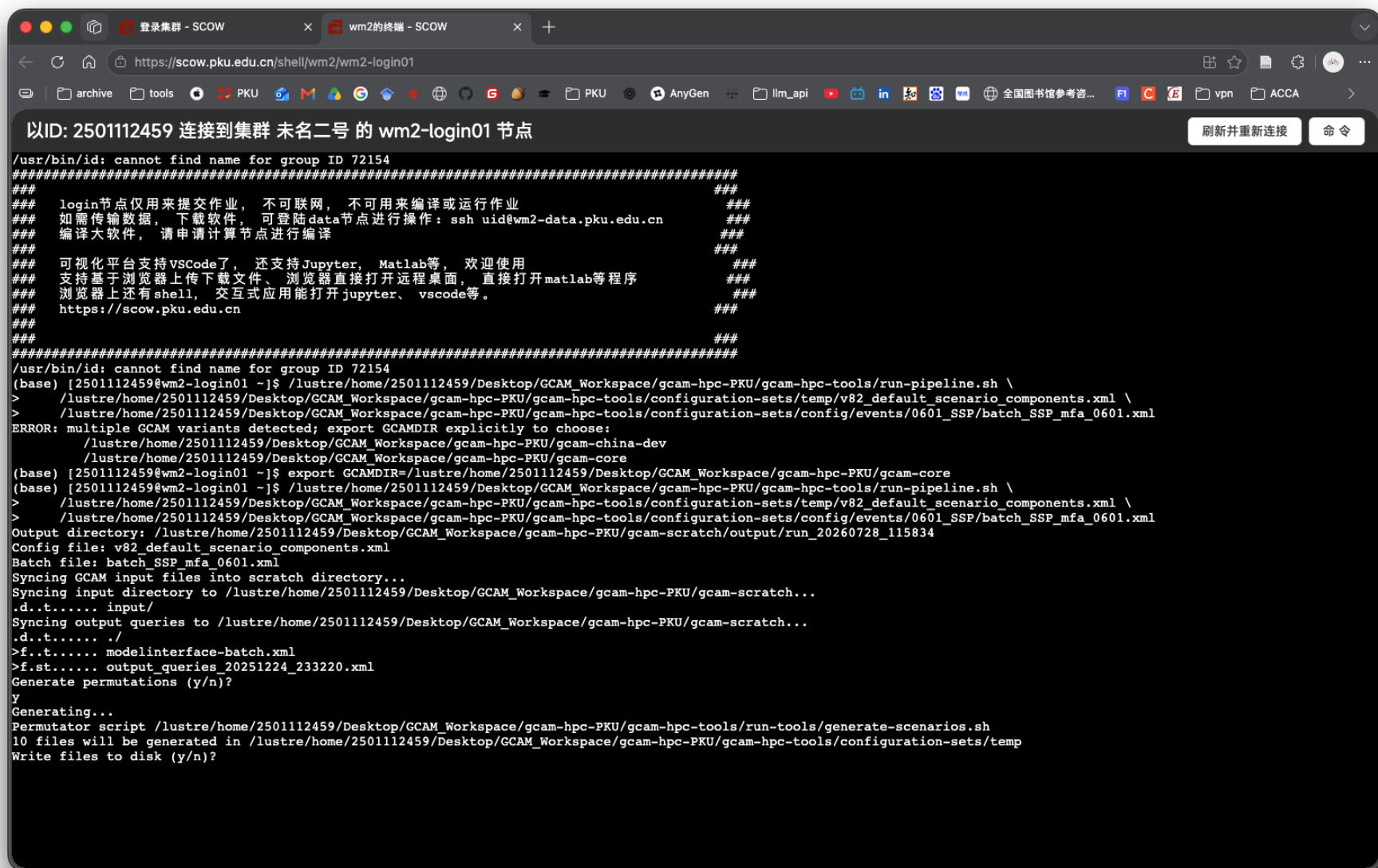


The screenshot shows a web browser window with a terminal interface. The browser's address bar displays `https://scow.pku.edu.cn/shell/wm2/wm2-login01`. The terminal window has a title bar with two tabs: "登录集群 - SCOW" and "wm2的终端 - SCOW". Below the title bar, a message reads: "以ID: 2501112459 连接到集群 未名二号的 wm2-login01 节点". To the right of this message are two buttons: "刷新并重新连接" and "命令".

The terminal output shows the following sequence of commands and responses:

```
/usr/bin/id: cannot find name for group ID 72154
#####
### login节点仅用来提交作业，不可联网，不可用来编译或运行作业
### 如需传输数据，下载软件，可登陆data节点进行操作：ssh uid@wm2-data.pku.edu.cn
### 编译大软件，请申请计算节点进行编译
###
### 可视化平台支持VSCode了，还支持Jupyter，Matlab等，欢迎使用
### 支持基于浏览器上传下载文件、浏览器直接打开远程桌面，直接打开matlab等程序
### 浏览器上还有shell，交互式应用能打开jupyter、vscode等。
### https://scow.pku.edu.cn
###
#####
/usr/bin/id: cannot find name for group ID 72154
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
ERROR: multiple GCAM variants detected; export GCAMDIR explicitly to choose:
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-china-dev
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ export GCAMDIR=/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
Output directory: /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch/output/run_20260728_115834
Config file: v82_default_scenario_components.xml
Batch file: batch_SSP_mfa_0601.xml
Syncing GCAM input files into scratch directory...
Syncing input directory to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d..t..... input/
Syncing output queries to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d..t..... ./
>f..t..... modelinterface-batch.xml
>f.st..... output_queries_20251224_233220.xml
Generate permutations (y/n)?
y
```

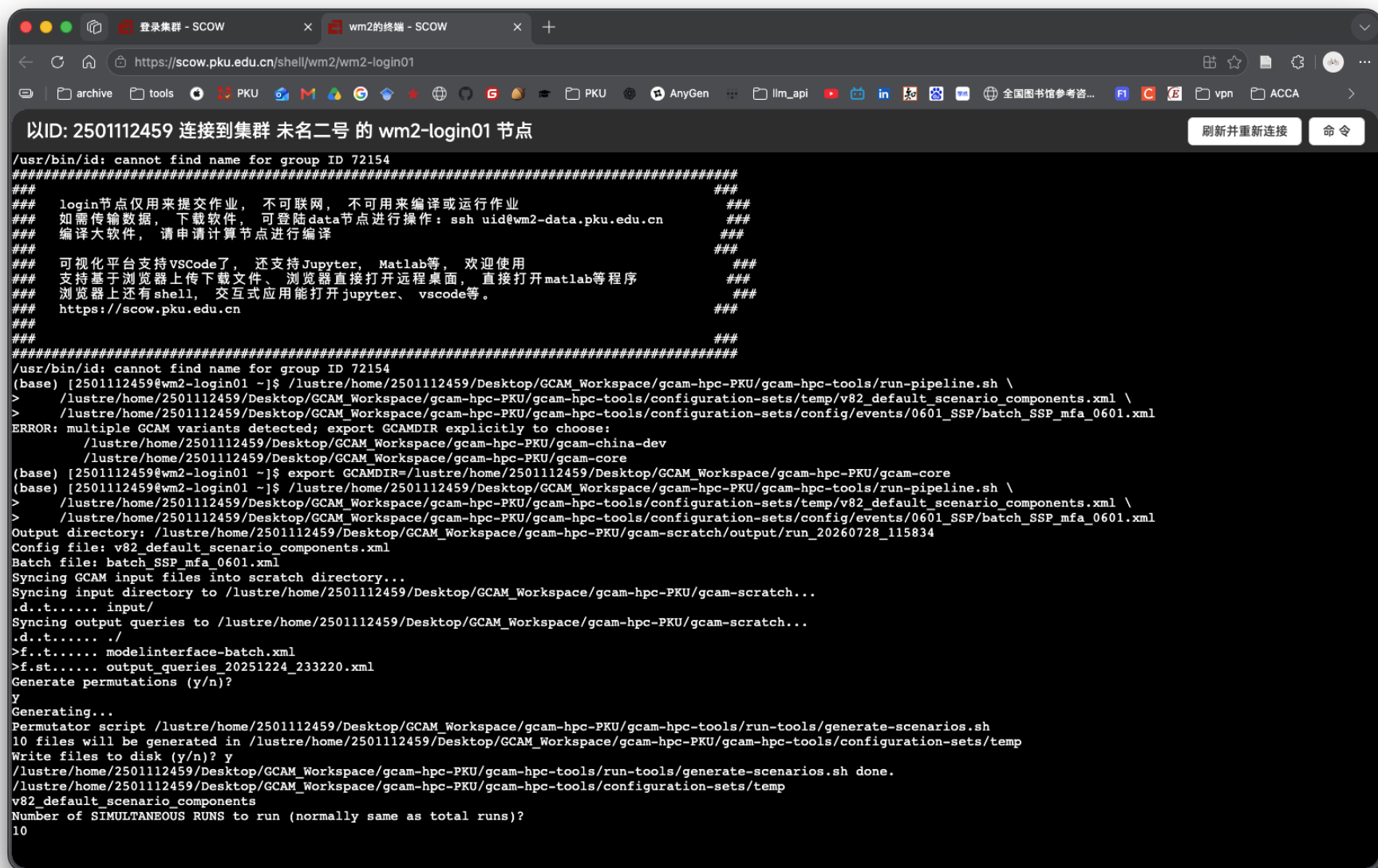
2--RUN: 实操演示



```
以ID: 2501112459 连接到集群 未名二号的 wm2-login01 节点 刷新并重新连接 命令

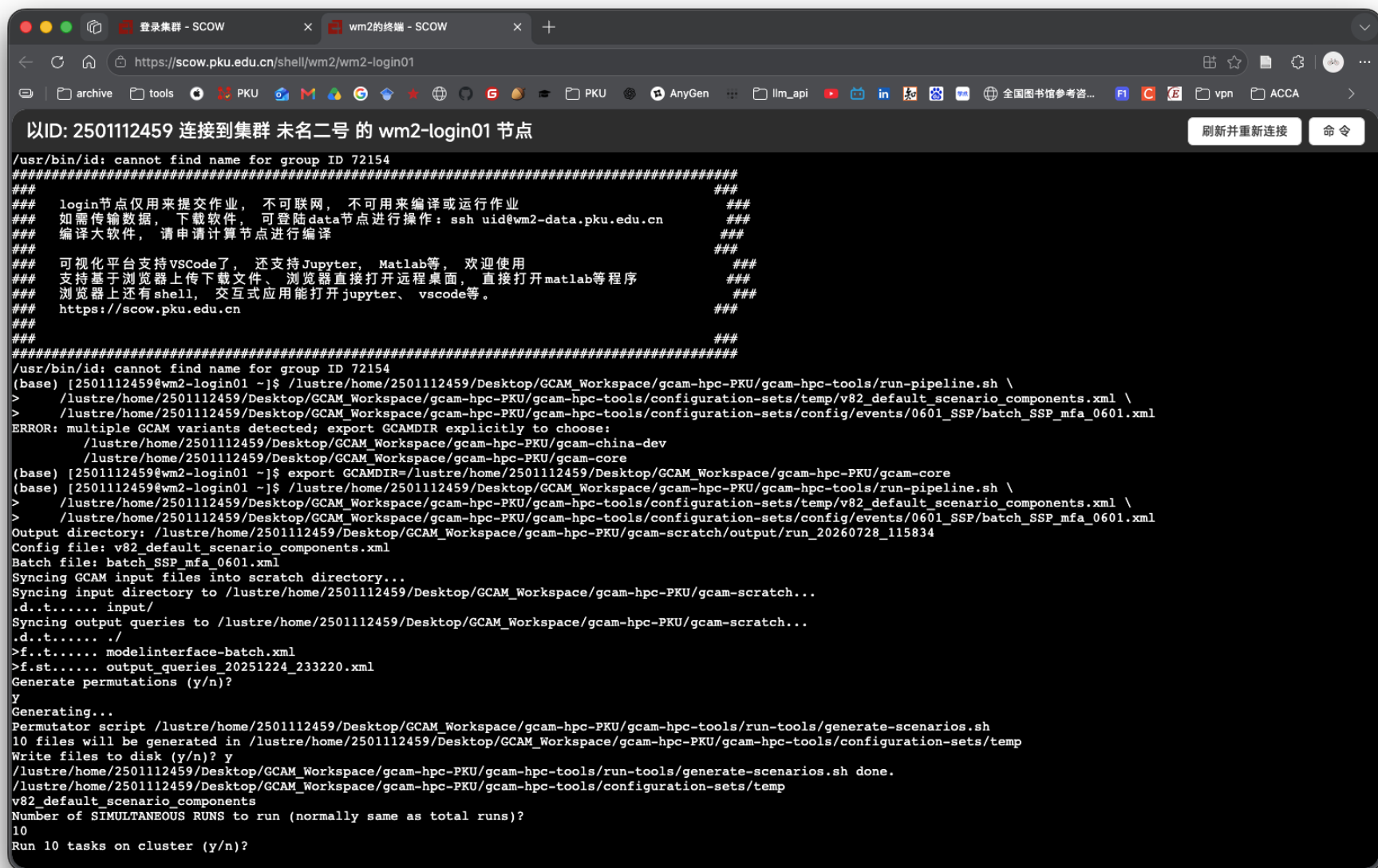
/usr/bin/id: cannot find name for group ID 72154
#####
###
### login节点仅用来提交作业，不可联网，不可用来编译或运行作业
### 如需传输数据，下载软件，可登陆data节点进行操作：ssh uid@wm2-data.pku.edu.cn
### 编译大软件，请申请计算节点进行编译
###
### 可视化平台支持VSCode了，还支持Jupyter，Matlab等，欢迎使用
### 支持基于浏览器上传下载文件、浏览器直接打开远程桌面，直接打开matlab等程序
### 浏览器上还有shell，交互式应用能打开jupyter、vscode等。
### https://scow.pku.edu.cn
###
#####
/usr/bin/id: cannot find name for group ID 72154
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
ERROR: multiple GCAM variants detected; export GCAMDIR explicitly to choose:
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-china-dev
    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ export GCAMDIR=/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
Output directory: /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch/output/run_20260728_115834
Config file: v82_default_scenario_components.xml
Batch file: batch_SSP_mfa_0601.xml
Syncing GCAM input files into scratch directory...
Syncing input directory to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d.t..... input/
Syncing output queries to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
.d.t..... ./
>f.t..... modelinterface-batch.xml
>f.st..... output_queries_20251224_233220.xml
Generate permutations (y/n)?
y
Generating...
Permutator script /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-tools/generate-scenarios.sh
10 files will be generated in /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp
Write files to disk (y/n)?
```

2--RUN: 实操演示



```
/usr/bin/id: cannot find name for group ID 72154
#####
###
### login节点仅用来提交作业，不可联网，不可用来编译或运行作业
### 如需传输数据，下载软件，可登陆data节点进行操作：ssh uid@wm2-data.pku.edu.cn
### 编译大软件，请申请计算节点进行编译
###
### 可视化平台支持VSCode了，还支持Jupyter，Matlab等，欢迎使用
### 支持基于浏览器上传下载文件、浏览器直接打开远程桌面，直接打开matlab等程序
### 浏览器上还有shell，交互式应用能打开jupyter、vscode等。
### https://scow.pku.edu.cn
###
#####
/usr/bin/id: cannot find name for group ID 72154
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
ERROR: multiple GCAM variants detected; export GCAMDIR explicitly to choose:
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    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ export GCAMDIR=/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
Output directory: /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch/output/run_20260728_115834
Config file: v82_default_scenario_components.xml
Batch file: batch_SSP_mfa_0601.xml
Syncing GCAM input files into scratch directory...
Syncing input directory to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
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Write files to disk (y/n)? y
/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-tools/generate-scenarios.sh done.
/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp
v82_default_scenario_components
Number of SIMULTANEOUS RUNS to run (normally same as total runs)?
10
```

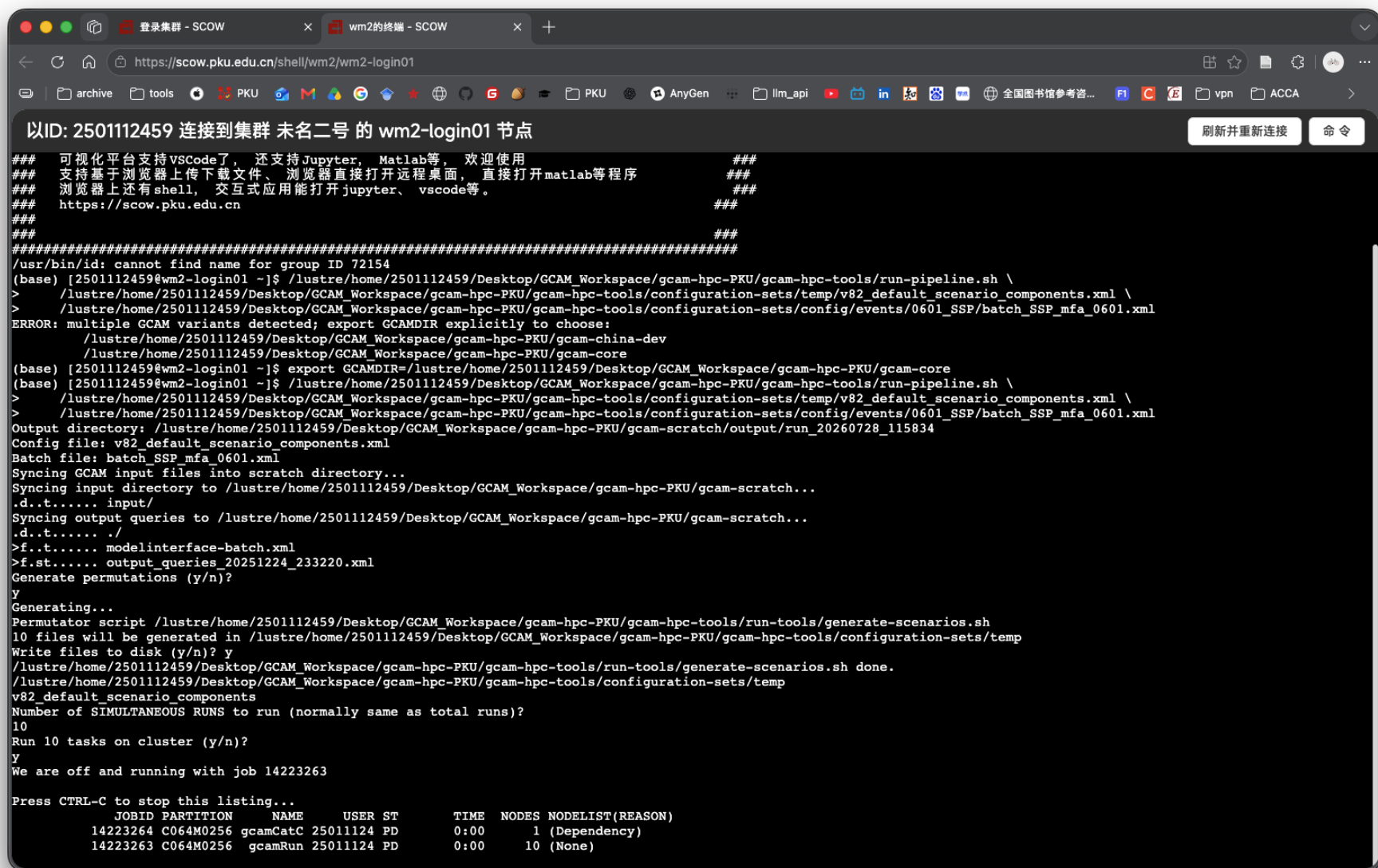
2--RUN: 实操演示



The screenshot shows a terminal window titled "wm2的终端 - SCOW" with a URL bar indicating the connection to "https://scow.pku.edu.cn/shell/wm2/wm2-login01". The terminal output shows the user logging in and running a script to generate scenarios. The script prompts for various configuration options and finally generates 10 files.

```
/usr/bin/id: cannot find name for group ID 72154
#####
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    /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ export GCAMDIR=/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
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> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/config/events/0601_SSP/batch_SSP_mfa_0601.xml
Output directory: /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch/output/run_20260728_115834
Config file: v82_default_scenario_components.xml
Batch file: batch_SSP_mfa_0601.xml
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.d.t..... input/
Syncing output queries to /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch...
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>f.st..... output_queries_20251224_233220.xml
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Permutator script /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-tools/generate-scenarios.sh
10 files will be generated in /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp
Write files to disk (y/n)? y
/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-tools/generate-scenarios.sh done.
/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp
v82_default_scenario_components
Number of SIMULTANEOUS RUNS to run (normally same as total runs)?
10
Run 10 tasks on cluster (y/n)?
```

2--RUN: 实操演示



```
### 可视化平台支持VSCode了, 还支持Jupyter, Matlab等, 欢迎使用
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ERROR: multiple GCAM variants detected; export GCAMDIR explicitly to choose:
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(base) [2501112459@wm2-login01 ~]$ export GCAMDIR=/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-core
(base) [2501112459@wm2-login01 ~]$ /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-pipeline.sh \
> /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp/v82_default_scenario_components.xml \
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Output directory: /lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-scratch/output/run_20260728_115834
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Write files to disk (y/n)? y
/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/run-tools/generate-scenarios.sh done.
/lustre/home/2501112459/Desktop/GCAM_Workspace/gcam-hpc-PKU/gcam-hpc-tools/configuration-sets/temp
v82_default_scenario_components
Number of SIMULTANEOUS RUNS to run (normally same as total runs)?
10
Run 10 tasks on cluster (y/n)?
y
We are off and running with job 14223263

Press CTRL-C to stop this listing...
JOBID PARTITION NAME USER ST TIME NODES NODELIST(REASON)
14223264 C064M0256 gcamCatC 25011124 PD 0:00 1 (Dependency)
14223263 C064M0256 gcamRun 25011124 PD 0:00 10 (None)
```

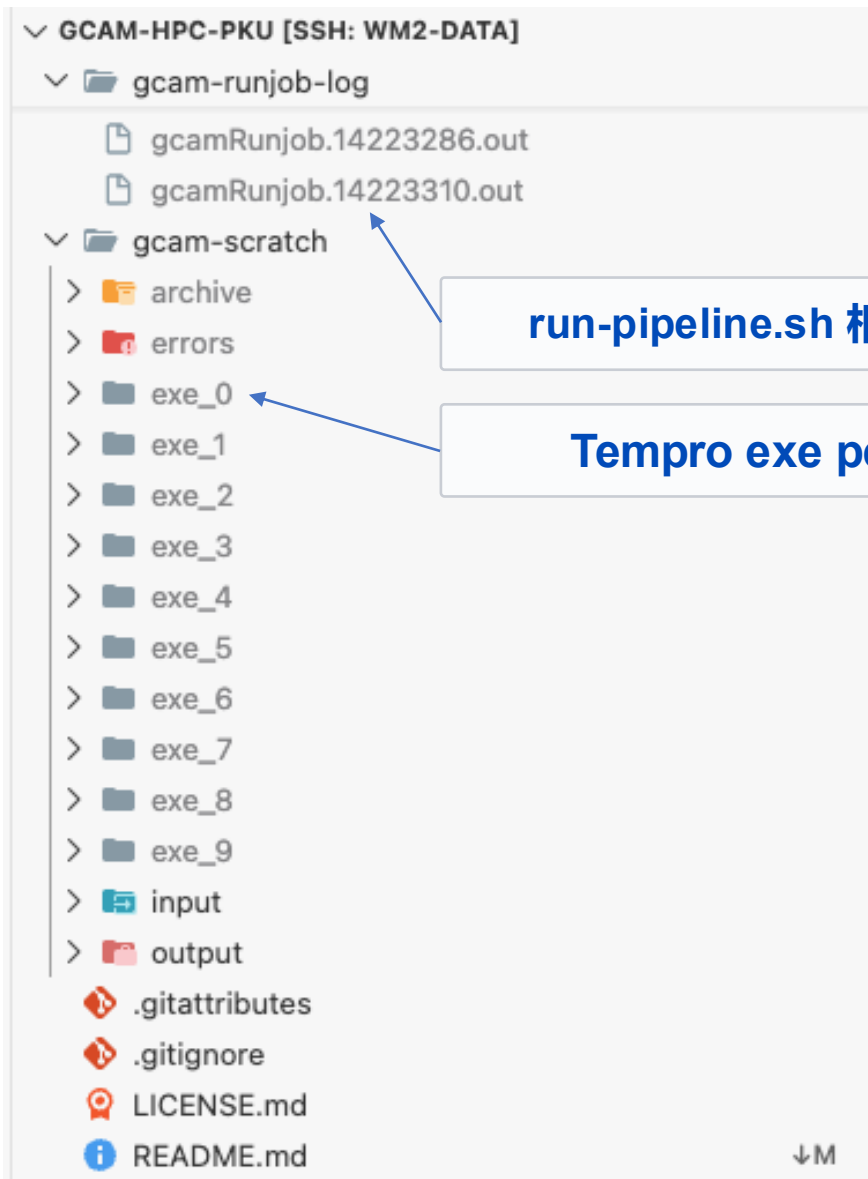
2--RUN: 实操演示

The screenshot displays the SCOW web interface in a browser window. The address bar shows the URL `https://scow.pku.edu.cn/jobs/runningJobs`. The page title is "未结束的作业 - SCOW". The left sidebar contains navigation links: "仪表盘", "作业", "登录集群", "应用", "文件管理", and "平台问答助手". The main content area is titled "未结束的作业" and features a search bar with "集群: 未名二号" and "作业ID:" fields, along with "搜索" and "刷新" buttons. Below the search bar is a table of running jobs.

作业ID	作业名	账户	分区	QOS	节点数	CPU核心数	加速卡卡数	状态	操作
14223311	gcamCatCSV	hpc2306191132	C064M0256G	normal	1	8	0	PENDING	  
14223310	gcamRun	hpc2306191132	C064M0256G	normal	10	80	0	RUNNING	  

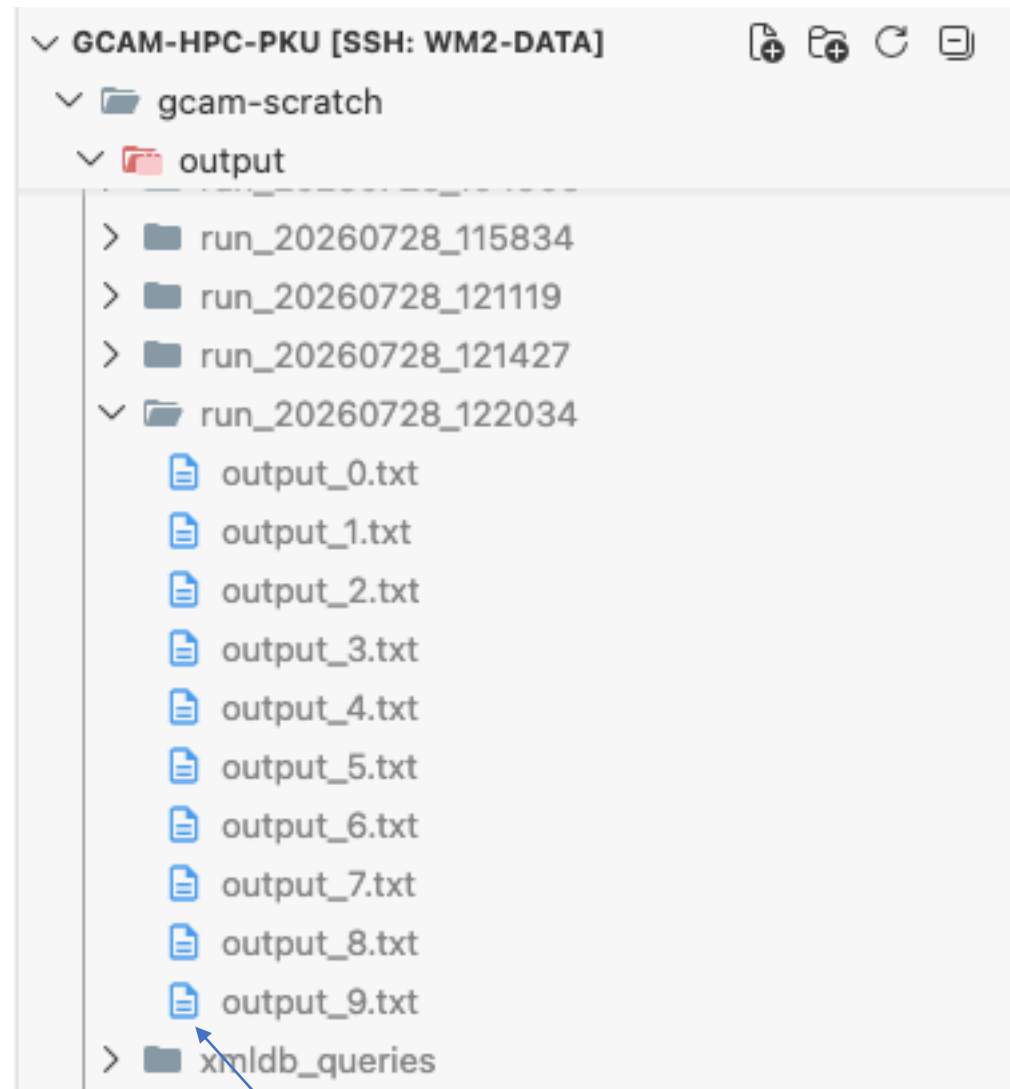
At the bottom right of the table, there is a pagination control showing "1" and "50 条/页".

2--RUN: 实操演示



run-pipeline.sh 相关的error

Tempo exe per scenario



gcam related error

其他须知

常见问题

- 未解压 gcam-hpc-pku-libs.zip 内的 archive/libs.zip, 导致 build-tools/libs/ 不完整
- 工作区存在多个 gcam-* 源码目录, 或 GCAMDIR 指向错误
- cluster profile 选择错误, Slurm partition、module 或依赖路径不匹配
- RUN 前未生成 queries-*.xml, 或 modelinterface-batch.xml 仍指向旧文件名
- QUERY 的 region / query 名称与定义文件不一致

工具集与文档

仓库 <https://github.com/jerrysong0128/gcam-hpc-PKU>

Wiki Home · 1-Build · 2-Run · 3-Query

源码 gcam-core: JGCRI/gcam-core · gcam-china: umd-cgs/gcam-china

致谢 本仓库基于弗吉尼亚大学Jay Fuhrman的前期工作 (<https://github.com/jayfuhrman/gcam-hpc-starterpack>) 构建, 在此对表示感谢

三步速览

BUILD 上传源码 → 解压 libs → source environment.sh → make gcam

QUERY 选择 regions / queries → build_queries.py → 配置 modelinterface-batch.xml

RUN run-pipeline.sh → 每个情景运行 GCAM + QUERY → 自动合并 CSV

其他内容详见Wiki页面

Home
Jerry Song edited this page yesterday · [7 revisions](#)

gcam-hpc-PKU Wiki

Parallel Runs for Scenario Matrix Construction

(Compatible with any version of GCAM-Core/GCAM-China)

Sequential Execution:

Scenario A → Scenario B → → 2-8 hours

High-Performance Parallel Execution:

..... → Scenario A → (Merged into lightweight CSV files) → 20-30 minutes

本 Wiki 说明如何在北京大学高性能计算平台上完成 GCAM 的 BUILD → QUERY 配置 → RUN → CSV 合并流程。

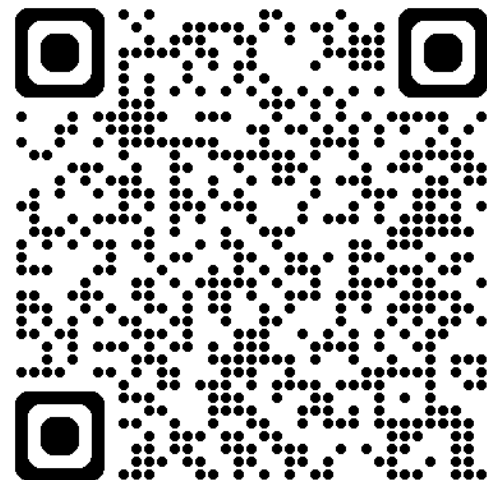
English: [Home-EN](#) | [Build-EN](#) | [Run-EN](#) | [Query-EN](#)

Clone this wiki locally
<https://github.com/jerrysong0128/>

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- 中文版
 - [首页](#)
 - [1. 编译 GCAM](#)
 - [2. 运行 GCAM](#)
 - [3. 查询输出](#)
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 - [Home](#)
 - [1. Build GCAM](#)
 - [2. Run GCAM](#)
 - [3. Query GCAM Output](#)

<https://github.com/jerrysong0128/gcam-hpc-PKU/wiki>



代码库建立在弗吉尼亚大学 Jay Fuhrman 先前工作的基础上
特此致谢

<https://github.com/jayfuhrman/gcam-hpc-starterpack>

GCAM-HPC-PKU

基于计算集群的GCAM工具

Jerry Song 宋京阳
jerrysong2025@stu.pku.edu.cn
2026年8月1日

